ODE Model with Life Contingency Application

Technical Model Report

1. Purpose and Scope

This report presents an ordinary differential equation model for a simplified life actuarial reserve problem. The model combines numerical ODE methods with life contingency assumptions so that reserve movement can be studied as a function of interest accumulation, premium inflow, mortality intensity, and death benefit exposure.

The model is not intended to be a production pricing or statutory valuation system. It is a controlled technical example that focuses on model structure, numerical validation, and actuarial interpretation. The simplified setting makes the relationship between the differential equation, the implementation, and the reserve results easier to inspect.

Model components.

- Numerical solvers: Euler, Heun, and fourth-order Runge-Kutta methods.
- Validation tests: closed-form ODE examples used to check solver behavior.
- Reserve model: a simplified continuous life reserve based on Thiele's differential equation.
- Scenario analysis: mortality, interest, and benefit sensitivities over a 30-year horizon.
- Outputs: reserve paths, validation diagnostics, and summary tables.

2. Model Structure

The project uses a modular structure. A model supplies a derivative function, an initial state, and a time grid. A numerical solver then advances the state across that grid. This design allows the same solver interface to be used for both theoretical validation examples and the actuarial reserve application.

Layer	**Role**	**Output**
Numerical engine	Implements Euler, Heun, and RK4 update rules.	State path over time.
Validation models	Use ODEs with analytic solutions, including exponential decay and logistic growth.	Maximum absolute error by solver.
Life contingency application	Uses Thiele's differential equation for a simplified continuous life reserve.	Reserve paths under selected assumptions.
Reporting layer	Writes CSV outputs and SVG/PDF figures.	Reproducible model outputs and diagnostics.

3. Numerical Methodology

The numerical component compares three standard one-step ODE methods. Euler's method is included as a simple baseline. Heun's method improves on Euler through a predictor-corrector step. Fourth-order Runge-Kutta is used as the main scenario solver because it usually provides much better accuracy for a given step size.

Method	Implementation role	Expected behavior
Euler	Baseline solver	Simple and transparent, but relatively high discretization error.
Heun	Improved Euler solver	Uses average slope information to reduce error.
RK4	Primary scenario solver	Higher-order method with materially lower error in validation tests.

Validation procedure. Before using the engine for the reserve model, the solvers are tested against two equations with known closed-form solutions. The error measure is the maximum absolute difference between the numerical path and the analytic path over the selected grid. This provides a compact check that the numerical methods are behaving as expected before they are applied to actuarial scenarios.

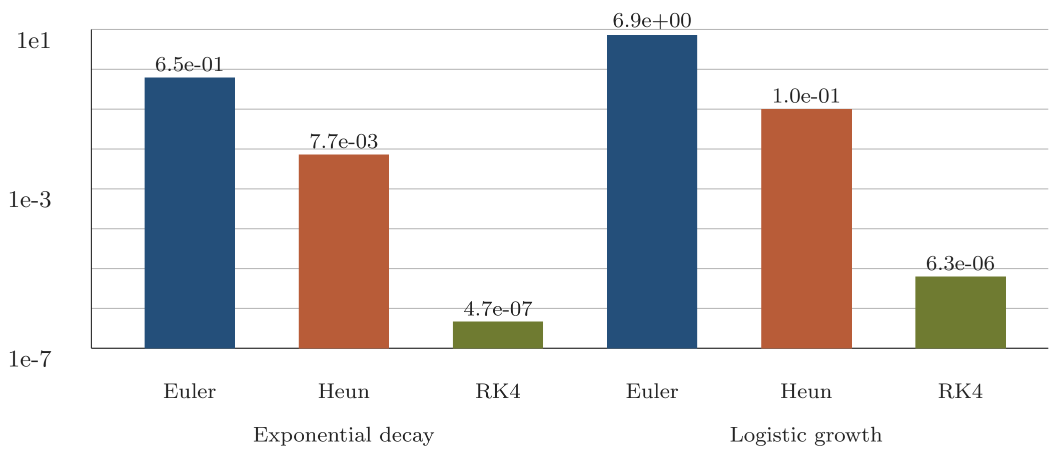

Figure 1: Solver validation against closed-form ODE solutions. Lower values indicate smaller maximum absolute error over the evaluation grid.

In these validation cases, the observed ranking is consistent with theory: RK4 has the smallest errors, Heun improves materially on Euler, and Euler remains useful mainly as a transparent baseline.

4. Life Contingency Application

The application layer applies the numerical engine to a simplified continuous life insurance reserve problem. The reserve is modeled as a function of interest accumulation, premium inflow, mortality intensity, and death benefit exposure. This connects the numerical model to a recognizable life contingency setting: valuing how future benefit obligations and financial assumptions affect the reserve held for a policy.

Reserve equation.

$$\frac{dV}{dt} = \delta V + P - \mu(B - V)$$

In this equation, V denotes the reserve, δ is the force of interest, P is the annual premium rate, μ is the mortality intensity, and B is the death benefit. The term δV reflects investment accumulation on the reserve; the premium term increases the reserve; the mortality term reflects expected claim pressure from death benefits net of reserve already held.

Input	Base value	Interpretation
δ	3.5%	Force of interest used for reserve accumulation.
μ	1.2%	Constant mortality intensity for the simplified model.
P	\$1,600	Annual premium rate entering the reserve equation.
B	\$100,000	Death benefit paid on claim.

Scenario design.

- Base case: standard assumptions for interest, mortality, premium, and benefit.
- Higher mortality: increases expected claim pressure and tests reserve sensitivity to mortality risk.
- Lower interest: reduces investment accumulation and tests sensitivity to financial assumptions.
- Higher benefit: increases death benefit exposure and tests sensitivity to policy design.

5. Results and Findings

The scenario results show how the reserve path changes when individual life contingency and financial assumptions are stressed. The base case produces positive reserve growth over the 30-year horizon. Higher mortality produces the lowest reserve path because expected claim pressure rises. Lower interest reduces accumulation, while a higher benefit reduces the reserve path through higher claim exposure.

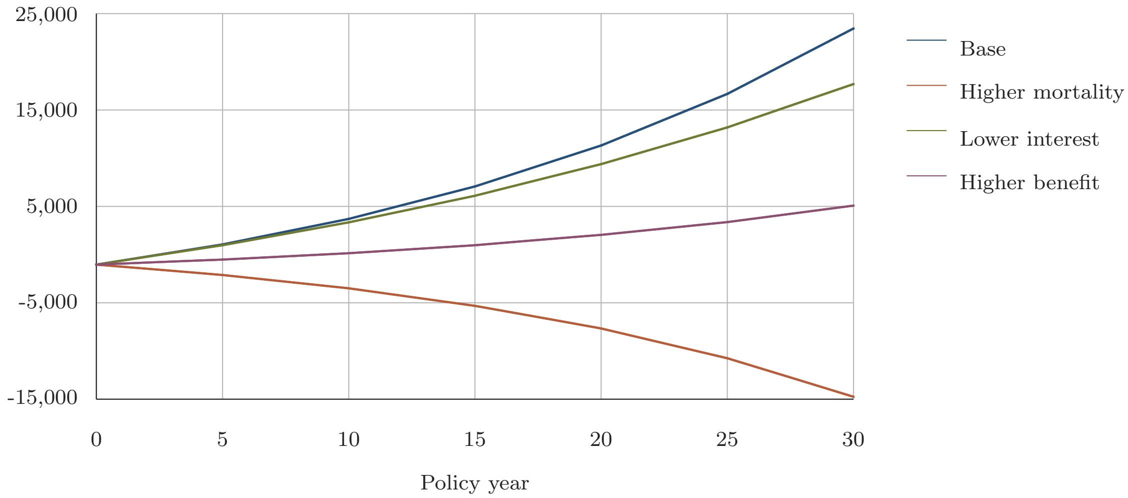

Figure 2: Reserve paths over a 30-year policy horizon under selected assumption sensitivities.

Scenario	30-year reserve	Finding
Base	$26,349	Premium and interest assumptions support reserve growth.
Higher mortality	$-14,731	Higher expected claim cost materially lowers the reserve path.
Lower interest	$20,146	Lower accumulation reduces the terminal reserve.
Higher benefit	$6,587	Larger benefit exposure lowers reserve growth relative to base.

Key findings.

- Validation results are consistent with the expected ranking of the numerical methods.
- The same ODE interface can be reused across theoretical examples and life reserve scenarios.
- The reserve application makes the output interpretable for life actuarial work because changes in assumptions have clear financial meaning.